

Quiz (Habanero)

- Q1.** Solve for t in the equation $2t + 5 = 11$.
(A) $t = 2$
(B) $t = 3$
(C) $t = 4$
(D) $t = 5$
(E) $t = 6$
- Q2.** Isolate x in the formula $x - 3 = 2y + 1$.
(A) $x = 2y + 4$
(B) $x = 2y - 4$
(C) $x = 3y + 1$
(D) $x = 3y - 1$
(E) $x = y + 3$
- Q3.** The temperature T in a city is given by $T = 2h - 5$, where h is the number of hours since sunrise. Solve for h .
(A) $h = \frac{T+5}{2}$
(B) $h = \frac{T-5}{2}$
(C) $h = 2T + 5$
(D) $h = 2T - 5$
(E) $h = T + 5$
- Q4.** The rainfall R in a region is related to the temperature T by the equation $R = 3T - 2$. Solve for T .
(A) $T = \frac{R+2}{3}$
(B) $T = \frac{R-2}{3}$
(C) $T = 3R + 2$
(D) $T = 3R - 2$
(E) $T = R + 2$
- Q5.** In a forest ecosystem, the population P of a species is given by $P = 2x - 5$, where x is the amount of food available. Solve for x .
(A) $x = \frac{P+5}{2}$
(B) $x = \frac{P-5}{2}$
(C) $x = 2P + 5$
(D) $x = 2P - 5$
(E) $x = P + 5$
- Q6.** Solve for y in the equation $y - 2 = 3x + 1$.
(A) $y = 3x + 3$
(B) $y = 3x - 3$
(C) $y = 2x + 1$
(D) $y = 2x - 1$
(E) $y = x + 2$
- Q7.** The water level W in a lake is given by $W = 2t + 1$, where t is the number of days since the last rainfall. Solve for t .
(A) $t = \frac{W-1}{2}$
(B) $t = \frac{W+1}{2}$
(C) $t = 2W + 1$
(D) $t = 2W - 1$
(E) $t = W - 1$
- Q8.** Isolate z in the equation $z + 2 = 2x - 3$.
(A) $z = 2x - 5$
(B) $z = 2x + 5$
(C) $z = x - 2$
(D) $z = x + 2$
(E) $z = 3x - 1$

Open Ended Questions (FRQ)

- Q9.** A city's average temperature T is related to the amount of rainfall R by the equation $T = 2R - 5$. Solve for R and explain your steps.
- Q10.** In a particular ecosystem, the population P of a species is given by $P = 3x - 2$, where x is the amount of food available. Solve for x and describe a real-world scenario where this equation would be useful.

Chef's Tips

- Emphasize the importance of isolating variables in algebraic equations.
- Encourage students to check their work by plugging their solutions back into the original equations.
- Remind students to read each question carefully and identify the variable they are solving for.